



南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

技术分析是否真的有效

Does technical analysis really work?



小组成员：李根 鲜俊昊
甘政 王悯汐



时间：2025年12月24日



目录

CONTENTS

01 研究背景

03 技术指标

02 形态策略

04 结论



目录

研究背景

形态策略

技术指标

结论

01

研究背景



目录

研究背景

形态策略

技术指标

结论

研究背景



技术分析通过研究价格形态和量价指标来预测未来走势，是金融市场中应用最广泛的分析工具之一。众多经典形态和量价指标被交易者视为判断买卖时机的关键依据。

然而，以“有效市场假说”为核心的现代金融理论对此提出了根本性质疑，认为这些基于历史信息模式无法持续产生超额收益。这一理论与实践之间的巨大鸿沟，使得技术分析的有效性成为一个长期悬而未决的核心争议。

为检验这一命题，本研究采用了量化与时序分析相结合的新思路：

- **K线形态**层面：利用**DTW算法**，将图形识别转化为可计算的相似度问题，构建基于K线形态的交易策略
- **技术指标**层面：运用**VAR模型**，分析技术指标与收益率预测之间的动态关系



目录

研究背景

形态策略

技术指标

结论

02

形态识别与量化策略： 基于DTW的图形匹配

- 2.1 DTW算法介绍
- 2.2 策略构建
- 2.3 回测结果

目录

研究背景

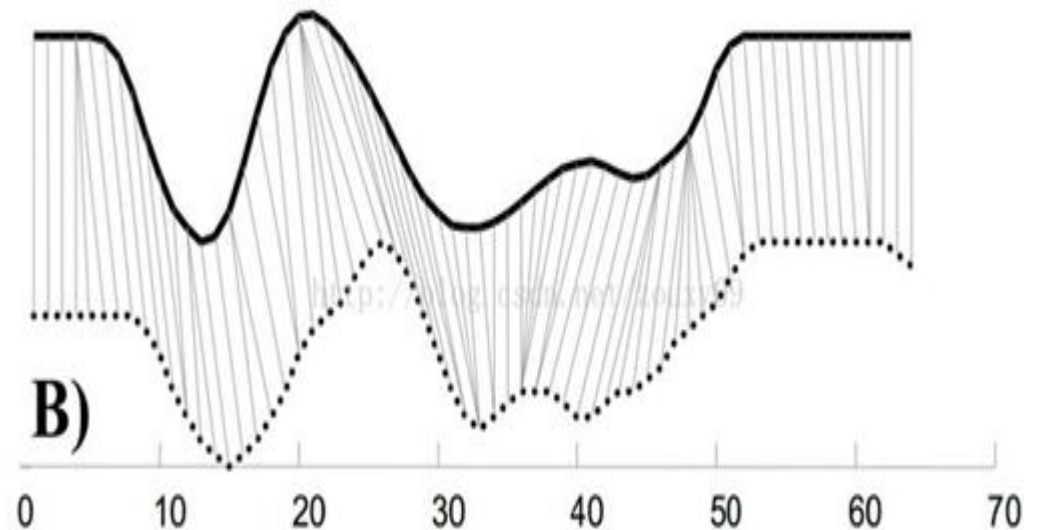
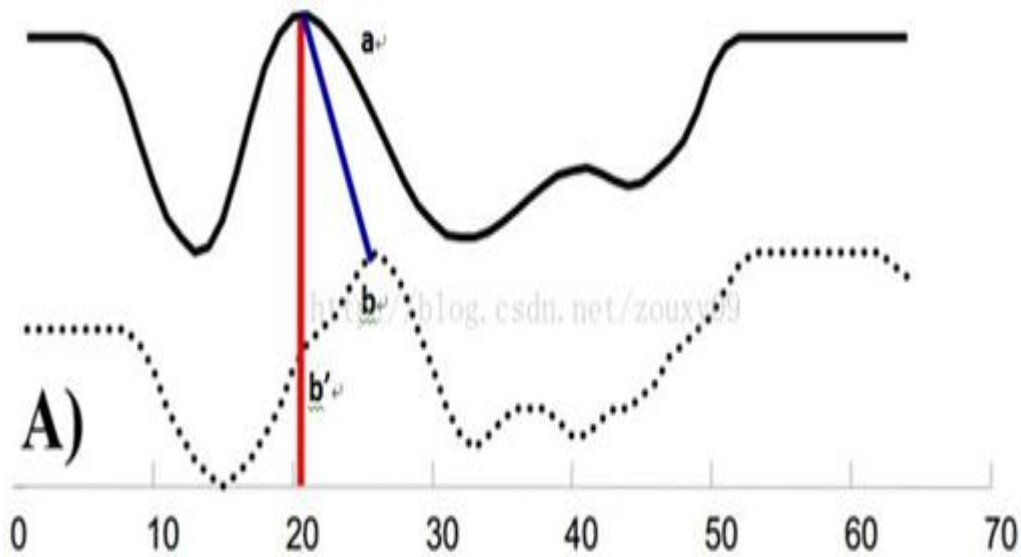
形态策略

技术指标

结论

2.1 DTW算法介绍

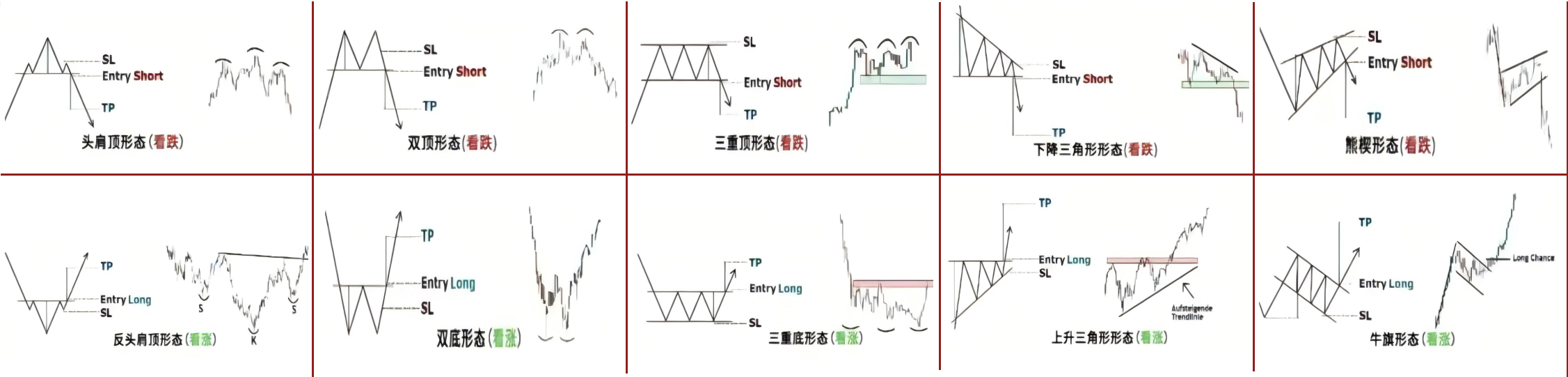
- 动态时间规整DTW(Dynamic Time Warping)
- 扭曲时间序列（即在时间轴上进行局部的缩放），使得两个序列的形态尽可能的一致，得到最大可能的相似度
- 计算两个时间序列的相似度，尤其适用于不同长度、不同节奏的时间序列
- 通过DTW算法来判断股价走势与经典K线形态的相似程度，进而决定买卖方向



2.2 策略构建

- 股票池：上证50成分股（50只股票）
- 时间范围：2020年10月1日至2025年9月30日
- 窗口长度：55天

策略所用的十大经典形态如下：



目录

研究背景

形态策略

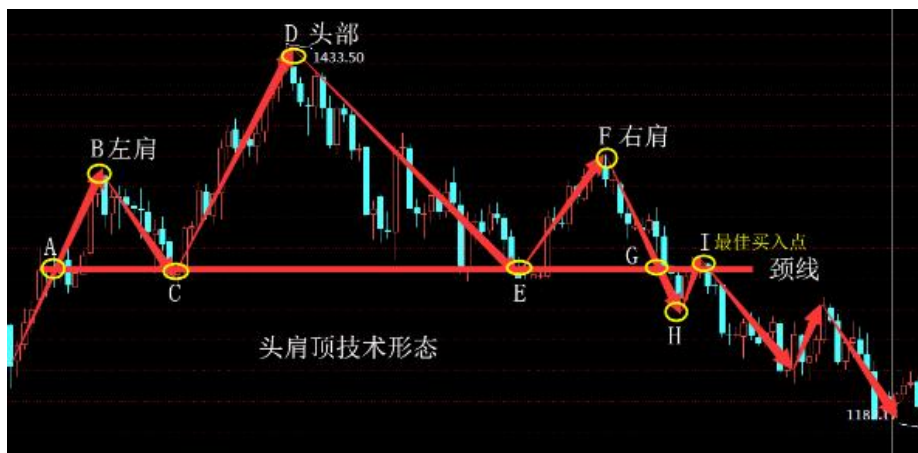
技术指标

结论

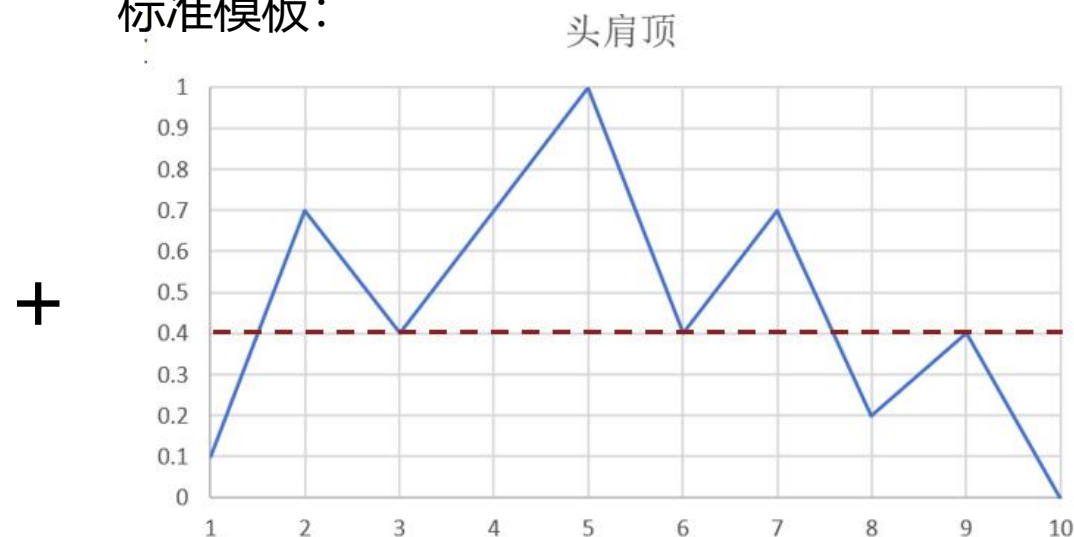
2.2 策略构建

- 针对十个技术分析的经典形态构建模板用于DTW相似度计算，通过DTW算法判别55天内每支股票股价k线图形态与这些标准模板形态的相似程度来决定是买入还是卖出
- 为防止主观臆断，设置了不同的相似度门槛和持有期
- 为防止持有期内出现形态反转导致策略亏损，在持有期内若识别到反向信号，则反向操作
- 为防止窗口期内重复识别到同一形态频繁操作，设置35天的信号冷却期

窗口期内K线形态（归一化后）：



标准模板：



- 计算DTW距离 < 设定门槛 (如0.1)
- 触发卖空信号，并持有等待反转信号或最长持有期结束 (如最长7天买入平仓)

2.3 回测结果



- 不同相似度门槛和持有期组合生成矩阵
- 交易次数矩阵、胜率矩阵、平均收益率矩阵（每次交易的平均收益率）
- 计算标准化后的收益矩阵，便于与基准收益(50只股票期间内平均每只的收益)对比
- 标准化收益 = $\frac{\text{交易次数} \times \text{平均收益率}}{50}$

表 1: 总样本 - 交易次数 (Transaction Frequency)									表 2: 总样本 - 胜率 (Win Rate)									表 3: 总样本 - 平均收益率 (Return Rate)								
DTW阈值	Day 1	Day 3	Day 5	Day 7	Day 9	Day 11	Day 13	Day 15	DTW阈值	Day 1	Day 3	Day 5	Day 7	Day 9	Day 11	Day 13	Day 15	DTW阈值	Day 1	Day 3	Day 5	Day 7	Day 9	Day 11	Day 13	Day 15
0.0500	58.0	58.0	58.0	58.0	58.0	58.0	58.0	58.0	0.0500	39.7	37.9	32.8	41.4	29.3	34.5	29.3	27.6	0.0500	-0.7725	-1.5327	-2.0904	-2.7011	-2.8695	-2.8591	-3.4817	-4.1180
0.0600	192.0	192.0	192.0	192.0	192.0	192.0	192.0	192.0	0.0600	47.9	39.6	43.2	47.9	45.3	42.7	43.8	43.8	0.0600	0.0009	-0.7503	-0.5454	-0.6808	-0.3847	-0.7275	-0.7143	-1.0500
0.0700	433.0	433.0	433.0	433.0	433.0	433.0	433.0	433.0	0.0700	47.8	52.2	47.6	48.0	49.4	50.1	50.1	50.8	0.0700	0.1275	0.2343	0.1214	-0.1115	-0.0274	-0.0682	-0.1485	-0.0814
0.0800	733.0	733.0	733.0	733.0	733.0	733.0	733.0	733.0	0.0800	51.6	52.0	50.1	51.6	49.8	48.4	52.0	50.6	0.0800	0.1820	0.1665	0.2278	0.0746	0.2239	0.1893	0.1363	0.0212
0.0900	994.0	994.0	994.0	994.0	994.0	994.0	994.0	994.0	0.0900	48.7	52.0	48.6	49.1	50.2	48.1	49.8	49.3	0.0900	0.0088	0.1273	-0.0568	-0.0571	-0.0620	-0.2913	-0.5246	-0.4947
0.1000	1200.0	1200.0	1200.0	1200.0	1200.0	1200.0	1200.0	1200.0	0.1000	48.8	50.7	50.2	49.1	50.5	48.8	48.7	49.0	0.1000	0.0740	0.1240	0.0018	-0.0850	-0.0420	-0.2080	-0.2655	-0.4016
0.1200	1461.0	1461.0	1461.0	1461.0	1461.0	1461.0	1461.0	1461.0	0.1200	49.2	51.0	50.4	49.3	50.3	49.4	49.4	49.5	0.1200	0.0498	0.1434	0.1107	0.0035	0.1010	0.0069	-0.0946	-0.1879
0.1400	1619.0	1619.0	1619.0	1619.0	1619.0	1619.0	1619.0	1619.0	0.1400	49.0	51.4	51.5	49.7	51.2	50.2	49.5	49.9	0.1400	0.0856	0.1866	0.1208	0.0145	0.0753	-0.0514	-0.0943	-0.1277
0.1600	1732.0	1732.0	1732.0	1732.0	1732.0	1732.0	1732.0	1732.0	0.1600	48.7	51.6	51.4	50.8	51.7	51.7	50.6	50.5	0.1600	0.0665	0.2240	0.2550	0.2339	0.3079	0.2615	0.1615	0.0649
10.0	1759.0	1759.0	1759.0	1759.0	1759.0	1759.0	1759.0	1759.0	10.0	48.5	52.0	51.2	50.9	52.0	51.5	51.5	51.3	10.0	0.0900	0.2127	0.1954	0.1535	0.2612	0.2356	0.2217	0.1725



目录

研究背景

形态策略

技术指标

结论

2.3 回测结果

- 随着DTW相似度要求增大（DTW阈值降低），对K线形态要求越严格，满足要求的K线越少，触发买卖越少，交易次数减少
- 主要看较小DTW阈值的组合（代表严格执行策略），DTW过大时图形识别过松会随机买卖
- 如0.06-Day7的58，代表期间策略触发了58次（开仓次数）

表 1: 总样本 - 交易次数 (Transaction Frequency)

DTW阈值	Day 1	Day 3	Day 5	Day 7	Day 9	Day 11	Day 13	Day 15
0.0500	58.0	58.0	58.0	58.0	58.0	58.0	58.0	58.0
0.0600	192.0	192.0	192.0	192.0	192.0	192.0	192.0	192.0
0.0700	433.0	433.0	433.0	433.0	433.0	433.0	433.0	433.0
0.0800	733.0	733.0	733.0	733.0	733.0	733.0	733.0	733.0
0.0900	994.0	994.0	994.0	994.0	994.0	994.0	994.0	994.0
0.1000	1200.0	1200.0	1200.0	1200.0	1200.0	1200.0	1200.0	1200.0
0.1200	1461.0	1461.0	1461.0	1461.0	1461.0	1461.0	1461.0	1461.0
0.1400	1619.0	1619.0	1619.0	1619.0	1619.0	1619.0	1619.0	1619.0
0.1600	1732.0	1732.0	1732.0	1732.0	1732.0	1732.0	1732.0	1732.0
10.0	1759.0	1759.0	1759.0	1759.0	1759.0	1759.0	1759.0	1759.0

注：行代表设定的允许进行交易的DTW相似度的门槛值，这个门槛值越小，代表越严格的按照模板形态进行交易
列代表持有期



目录

研究背景

形态策略

技术指标

结论

2.3 回测结果

- 整体看，随着DTW相似度要求增大（距离门槛值降低），有胜率降低的趋势
- 如0.06-Day7的41.4，代表期间58次交易中41.4%(即24次)的交易收益率为正（判断了正确方向）

表 2: 总样本 - 胜率 (Win Rate)

DTW阈值	Day 1	Day 3	Day 5	Day 7	Day 9	Day 11	Day 13	Day 15
0.0500	39.7	37.9	32.8	41.4	29.3	34.5	29.3	27.6
0.0600	47.9	39.6	43.2	47.9	45.3	42.7	43.8	43.8
0.0700	47.8	52.2	47.6	48.0	49.4	50.1	50.1	50.8
0.0800	51.6	52.0	50.1	51.6	49.8	48.4	52.0	50.6
0.0900	48.7	52.0	48.6	49.1	50.2	48.1	49.8	49.3
0.1000	48.8	50.7	50.2	49.1	50.5	48.8	48.7	49.0
0.1200	49.2	51.0	50.4	49.3	50.3	49.4	49.4	49.5
0.1400	49.0	51.4	51.5	49.7	51.2	50.2	49.5	49.9
0.1600	48.7	51.6	51.4	50.8	51.7	51.7	50.6	50.5
10.0	48.5	52.0	51.2	50.9	52.0	51.5	51.5	51.3

2.3 回测结果



- 严格执行K线形态策略时，绝大多数触发的交易收益率为负
- 如0.06-Day7的-0.6808，代表期间58次交易中每次交易的平均收益为-0.6808%

表 3: 总样本 - 平均收益率 (Return Rate)

DTW阈值	Day 1	Day 3	Day 5	Day 7	Day 9	Day 11	Day 13	Day 15
0.0500	-0.7725	-1.5327	-2.0904	-2.7011	-2.8695	-2.8591	-3.4817	-4.1180
0.0600	0.0009	-0.7503	-0.5454	-0.6808	-0.3847	-0.7275	-0.7143	-1.0500
0.0700	0.1275	0.2343	0.1214	-0.1115	-0.0274	-0.0682	-0.1485	-0.0814
0.0800	0.1820	0.1665	0.2278	0.0746	0.2239	0.1893	0.1363	0.0212
0.0900	0.0088	0.1273	-0.0568	-0.0571	-0.0620	-0.2913	-0.5246	-0.4947
0.1000	0.0740	0.1240	0.0018	-0.0850	-0.0420	-0.2080	-0.2655	-0.4016
0.1200	0.0498	0.1434	0.1107	0.0035	0.1010	0.0069	-0.0946	-0.1879
0.1400	0.0856	0.1866	0.1208	0.0145	0.0753	-0.0514	-0.0943	-0.1277
0.1600	0.0665	0.2240	0.2550	0.2339	0.3079	0.2615	0.1615	0.0649
10.0	0.0900	0.2127	0.1954	0.1535	0.2612	0.2356	0.2217	0.1725



目录

研究背景

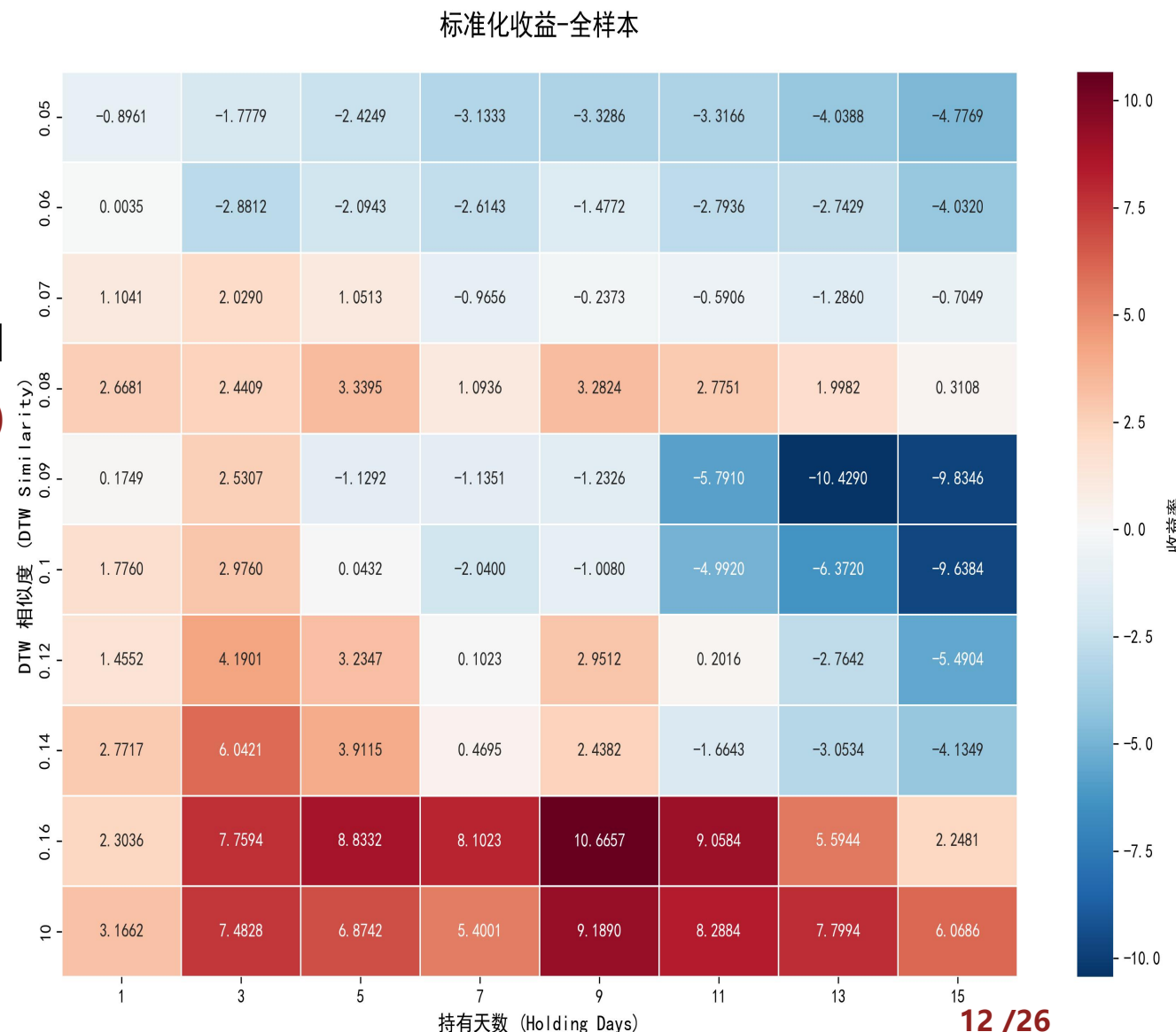
形态策略

技术指标

结论

2.3 回测结果

- 热力图如右
- 纵轴不同DTW相似度阈值
- 横轴为不同持有天数
- 用可与基准收益相比的标准化收益
- 在2020年10月1日至2025年9月30日期间，基准收益为54.97%(期间收益)
- 通过K线形态进行交易，收益率远低于基准收益
- 整体越严格地按K线形态进行买卖，有收益越差的趋势





目录

研究背景

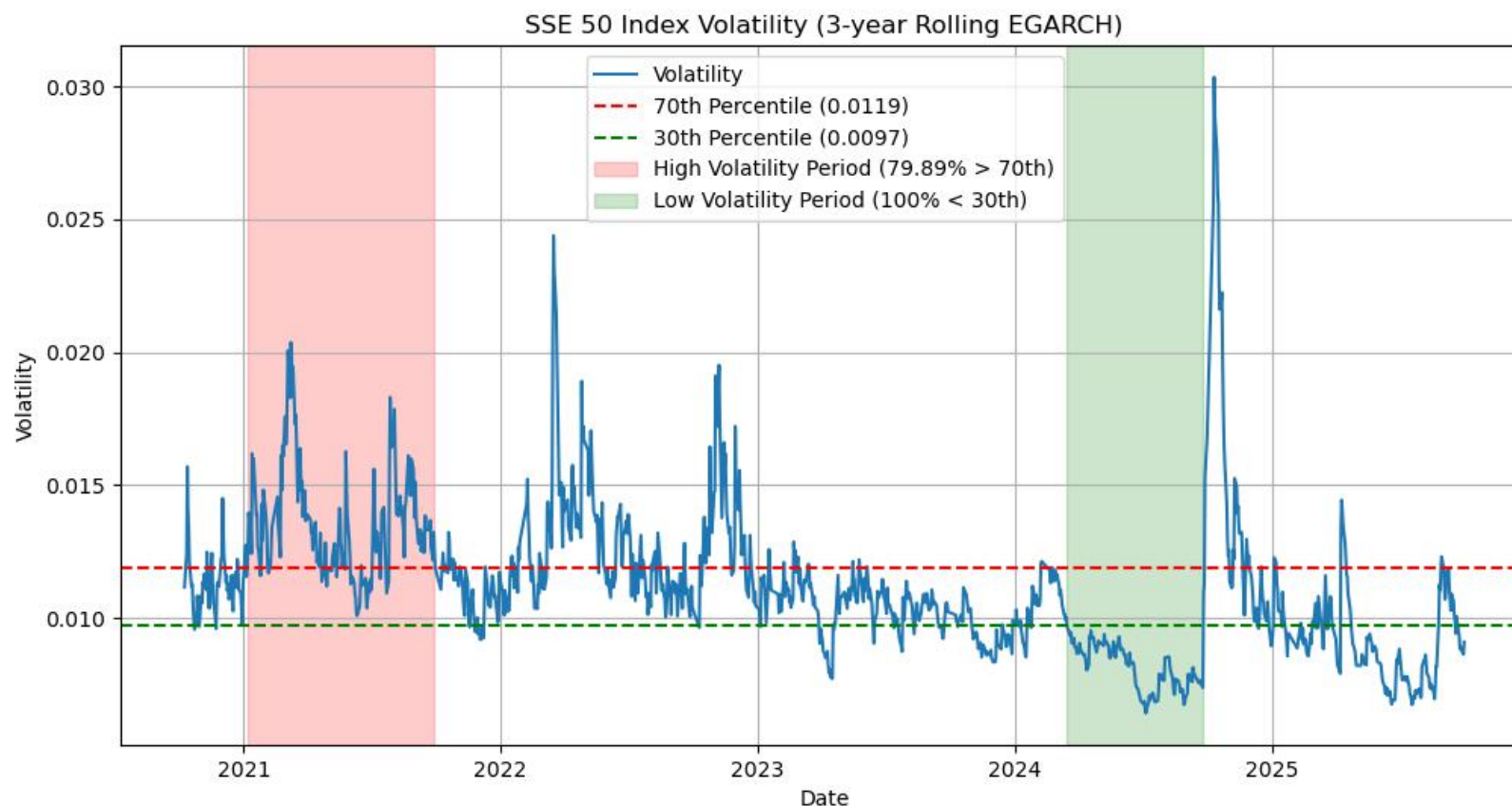
形态策略

技术指标

结论

高低波动期划分

- **方法：**基于EGARCH模型，估计上证50股指每天的波动率，采用750天滚动窗口
- **高波动期：**2021年1月6日-9月29日，78.89%样本高于70%分位数
- **低波动期：**2024年3月15日-9月24日，100%样本低于30%分位数



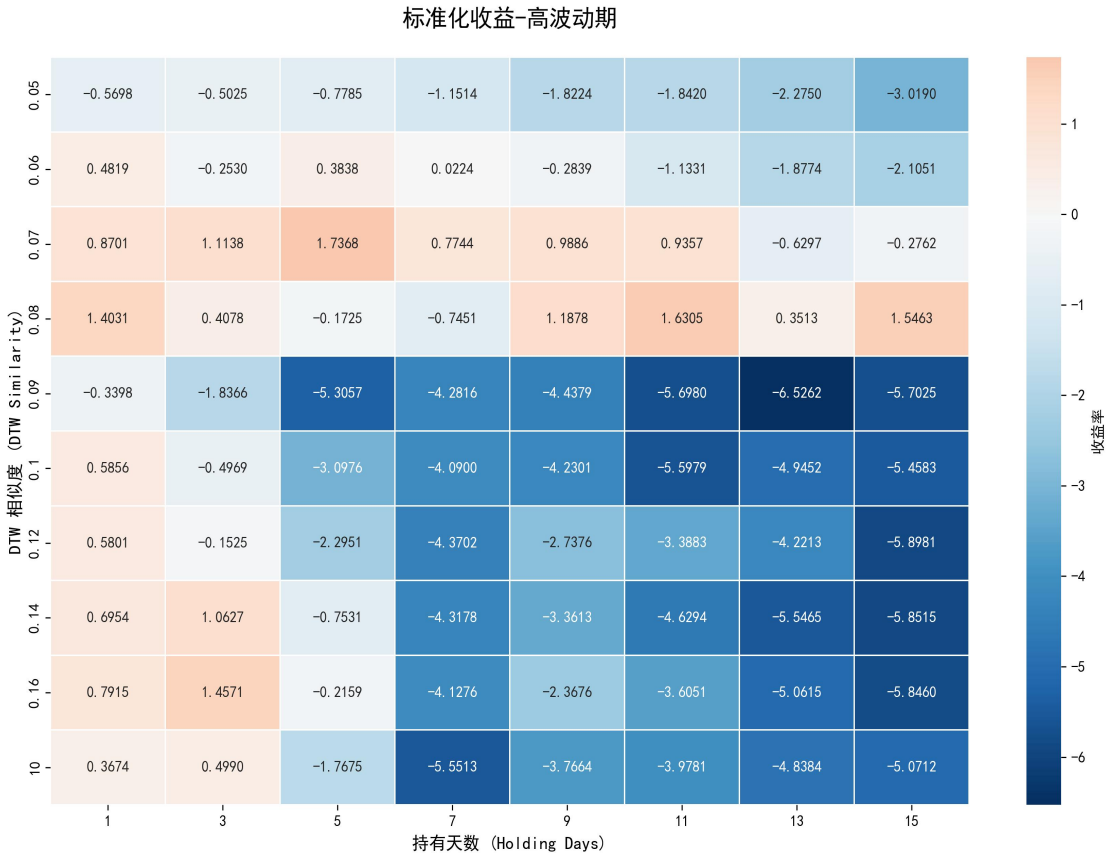
无论高波动期还是低波动期，K线策略均很难跑赢基准

研究背景

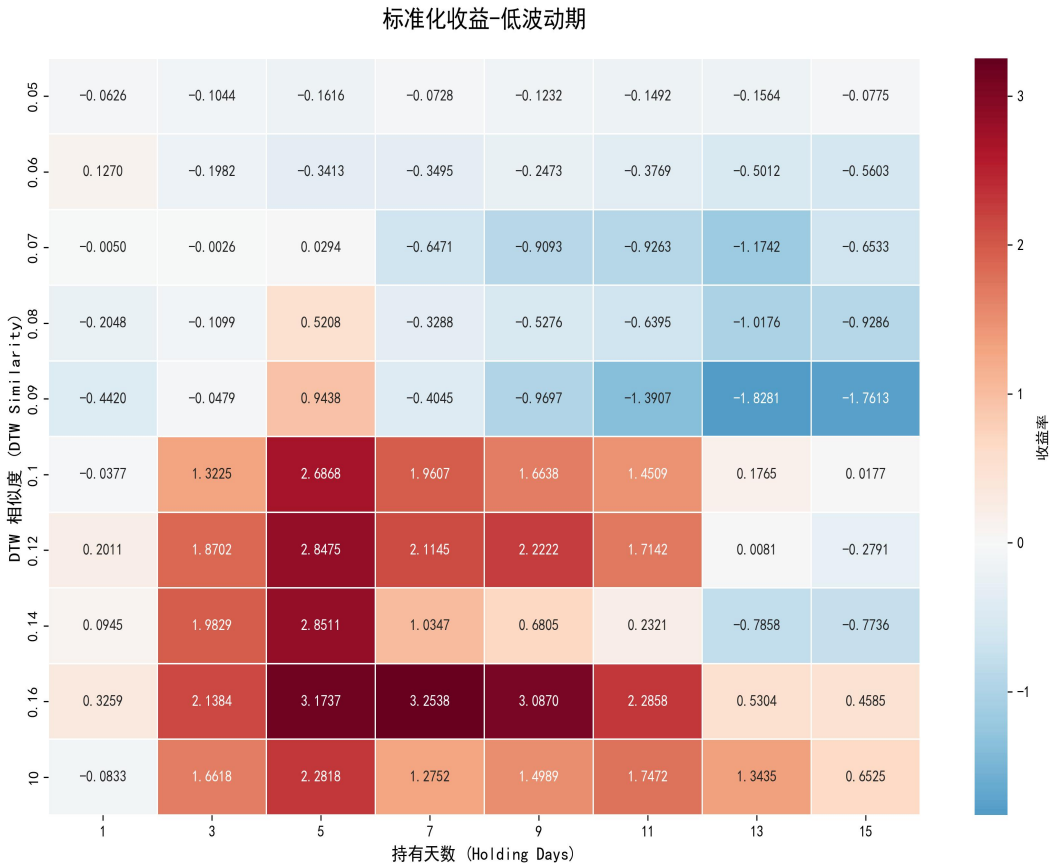
形态策略

技术指标

结论



基准收益：1.88%



基准收益：2.82%



目录

研究背景

形态策略

技术指标

结论

03

技术指标与收益率预测： 基于VAR的动态分析

- 3.1 数据来源与处理
- 3.2 数据检验
- 3.3 实证结果

3.1 数据来源与处理



- 数据选择：上证50指数日收盘价、成交量、最高价、最低价
- 数据来源：瑞思(RESSET)数据库
- 时间范围：2020年10月1日-2025年9月30日

指标名称	英文全称	指标类别	常用周期	主要用途	计算	方向
MACDI	Moving Average Convergence Divergence Histogram	趋势判断	(12,26,9)	识别趋势方向和转折点	• MACD线=快线EMA(12)-慢线EMA(26) • 信号线=EMA(MACD,9) • MACDI=MACD线-信号线	• 正值看涨 • 负值看跌
RSI	Relative Strength Index	反转信号	14日	判断超买超卖状态	• RS=平均上涨幅度/平均下跌幅度 • RSI=100-100/(1+RS)	• 50为中线 • 30以下强烈看涨 • 70以上强烈看跌
ROC	Rate of Change	动量强度	10日	测量价格变化速度	$ROC = \frac{P_t - P_{t-10}}{P_{t-10}} \times 100$	• 正值看涨 • 负值看跌
MFI	Money Flow Index	资金流量	14日	结合价格和成交量的动量指标，判断超买超卖	• 典型价格=(最高价+最低价+收盘价)/3 • 原始资金流=典型价格×成交量 • 根据收盘价区分每日正负资金流 • MFI=正资金流之和/正负资金流之和×100	• 50为中线 • 20以下强烈看涨 • 80以上强烈看跌



3.2 数据检验——散点图

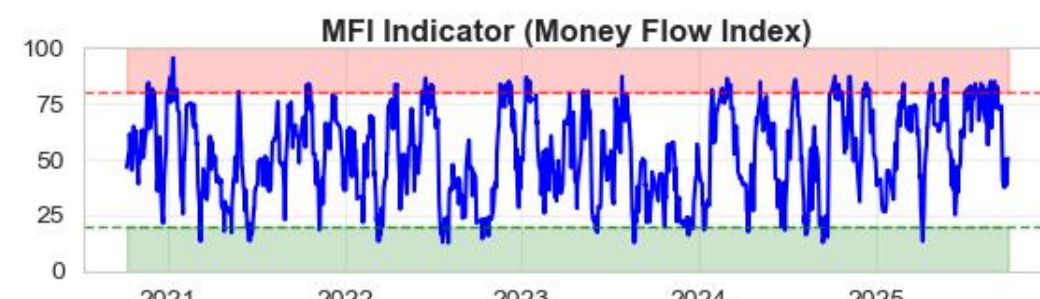
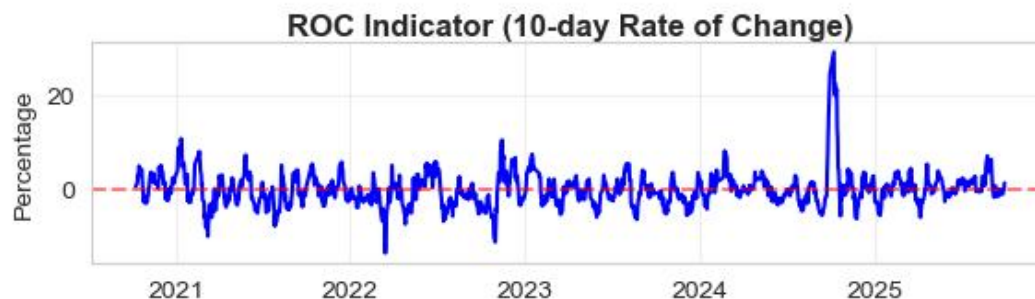
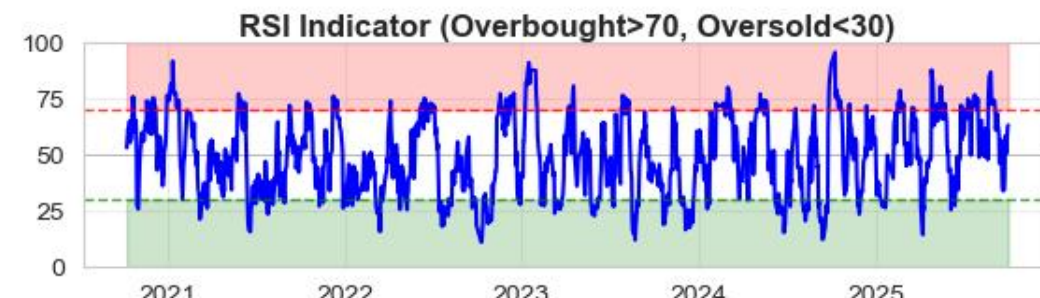
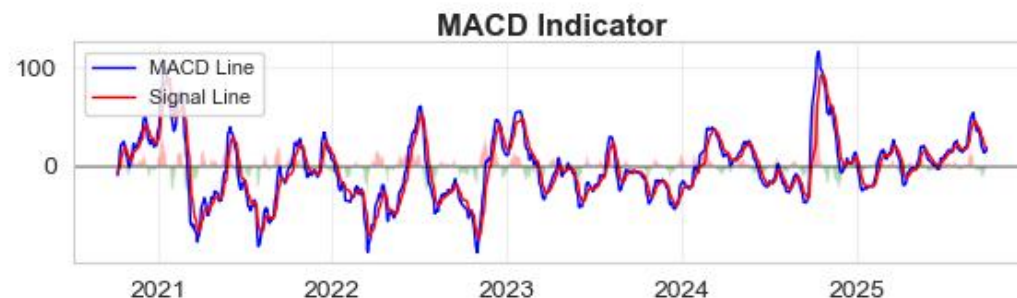
目录

研究背景

形态策略

技术指标

结论



研究背景

形态策略

技术指标

结论

变量	样本数	均值	标准差	偏度	峰度	25th	50th	75th
<i>log_return</i>	1212	-0.006	1.112	0.073	4.322	-0.580	-0.026	0.542
<i>MACDI</i>	1212	0.077	10.466	0.575	4.442	-5.863	0.128	5.798
<i>RSI</i>	1212	49.236	15.847	0.059	-0.531	38.089	48.729	60.827
<i>ROC</i>	1212	-0.025	3.543	1.603	9.900	-2.145	-0.198	1.778
<i>MFI</i>	1212	51.691	14.681	-0.093	-0.472	41.713	51.412	62.604



3.2 数据检验——平稳性检验

目录

研究背景

形态策略

技术指标

结论

序列	检验类型	ADF检验统计量	临界值	P值	结论
<i>log_return</i>	(0,0,0)	-34.90569	-2.566873	0.0000	平稳
<i>MACDI</i>	(0,0,1)	-10.54886	-2.566875	0.0000	平稳
<i>RSI</i>	(C,0,14)	-6.106584	-3.435599	0.0000	平稳
<i>ROC</i>	(0,0,20)	-5.515384	-2.566905	0.0000	平稳
<i>MFI</i>	(C,0,15)	-5.666170	-3.435604	0.0000	平稳

- 注：检验形式中(C,T,K)分别表示检验模型中含有的常数项、趋势项和滞后阶数，其中K的选择以AIC值最小为准则，临界值和平稳性结论都是基于1%的显著性水平。



目录

研究背景

形态策略

技术指标

结论

3.3 实证结果——MACDI

- **格兰杰因果检验**: 对数收益率是MACDI的格兰杰原因, 但MACDI不是对数收益率的格兰杰原因
- **脉冲相应**: MACDI 一个单位的正向冲击, 对未来对数收益率无显著影响
- **方差分解**: MACDI的方差在很大程度上由股票收益率变化所解释

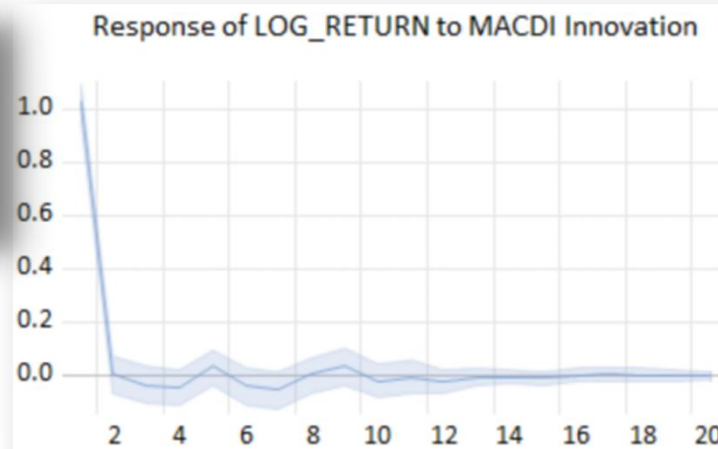
◆ VAR模型 (根据信息准则确定滞后阶数为10)

$$\begin{aligned} MACDI_{1t} &= \alpha_{10} + \alpha_{11}MACDI_{1,t-1} + \cdots + \alpha_{1k}MACDI_{1,t-k} + \beta_{11}LR_{1,t-1} + \cdots + \beta_{1k}LR_{1,t-k} + \varepsilon_{1t} \\ LR_{2t} &= \alpha_{20} + \alpha_{21}LR_{2,t-1} + \cdots + \alpha_{2k}LR_{2,t-k} + \beta_{11}MACDI_{2,t-1} + \cdots + \beta_{1k}MACDI_{2,t-k} + \varepsilon_{2t} \end{aligned}$$

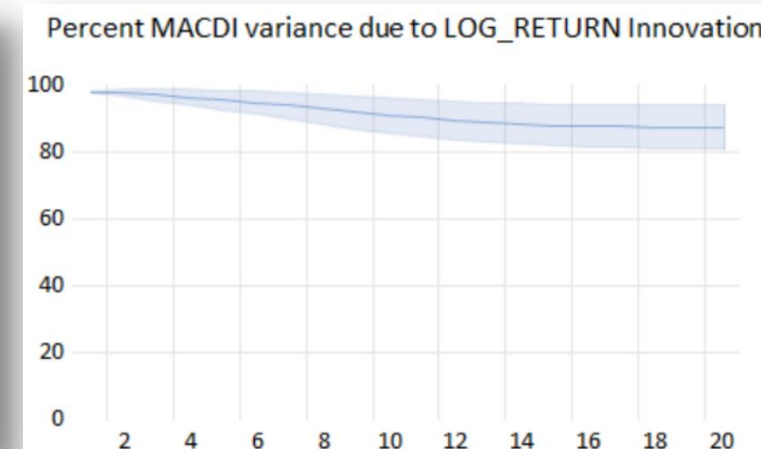
◆ 格兰杰因果检验

Pairwise Granger Causality Tests			
Date: 12/21/25 Time: 11:48			
Sample: 10/09/2020 9/30/2025			
Lags: 10			
Null Hypothesis:	Obs	F-Statistic	Prob.
MACDI does not Granger Cause LOG_RETURN	1202	1.53947	0.1199
LOG_RETURN does not Granger Cause MACDI		3.10892	0.0006

◆ 广义脉冲响应



◆ 方差分解





目录

研究背景

形态策略

技术指标

结论

3.3 实证结果——RSI

- **格兰杰因果检验**: 对数收益率是RSI的格兰杰原因, 但**RSI不是对数收益率的格兰杰原因**
- **脉冲相应**: RSI一个单位的正向冲击, **对未来对数收益率无显著影响**
- **方差分解**: RSI的方差在很大程度上由股票收益率变化所解释

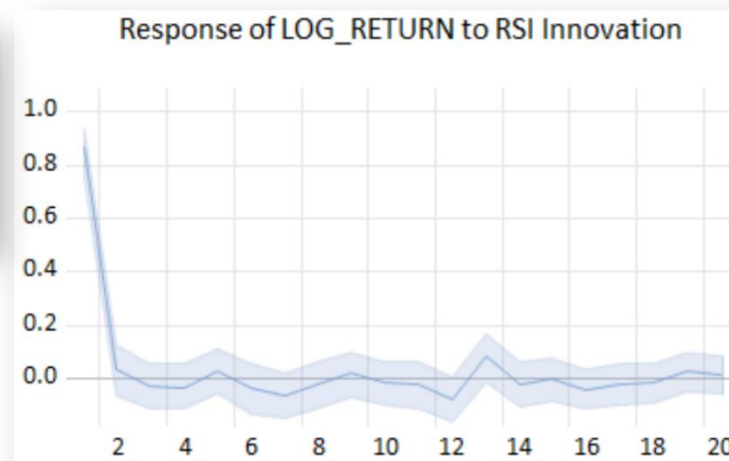
◆ VAR模型 (根据信息准则确定滞后阶数为14)

$$RSI_{1t} = \alpha_{10} + \alpha_{11}RSI_{1,t-1} + \cdots + \alpha_{1k}RSI_{1,t-k} + \beta_{11}LR_{1,t-1} + \cdots + \beta_{1k}LR_{1,t-k} + \varepsilon_{1t}$$
$$LR_{2t} = \alpha_{20} + \alpha_{21}LR_{2,t-1} + \cdots + \alpha_{2k}LR_{2,t-k} + \beta_{11}RSI_{2,t-1} + \cdots + \beta_{1k}RSI_{2,t-k} + \varepsilon_{2t}$$

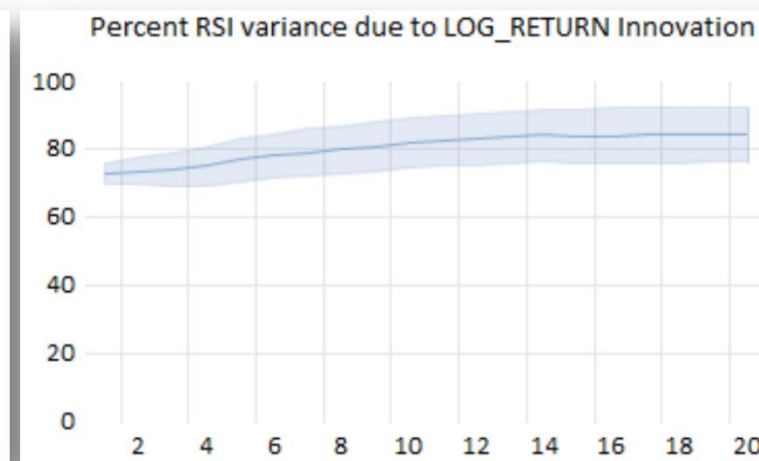
◆ 格兰杰因果检验

Pairwise Granger Causality Tests			
Date: 12/21/25 Time: 11:52			
Sample: 10/09/2020 9/30/2025			
Lags: 14			
Null Hypothesis:	Obs	F-Statistic	Prob.
RSI does not Granger Cause LOG_RETURN	1198	1.10970	0.3440
LOG_RETURN does not Granger Cause RSI		55.1206	5E-118

◆ 广义脉冲响应



◆ 方差分析





目录

研究背景

形态策略

技术指标

结论

3.3 实证结果——ROC

- 格兰杰因果检验：对数收益率是ROC的格兰杰原因，但ROC不是对数收益率的格兰杰原因
- 脉冲相应：ROC一个单位的正向冲击，对未来对数收益率无显著影响
- 方差分解：ROC的方差在很大程度上由股票收益率变化所解释

◆ VAR模型（根据信息准则确定滞后阶数为10）

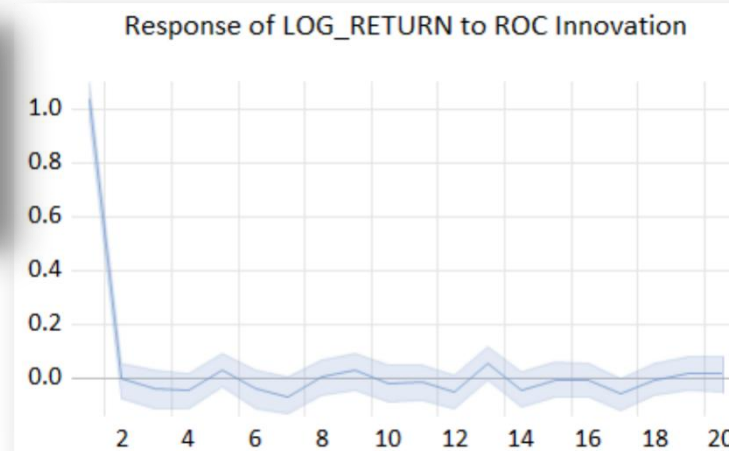
$$\begin{aligned} ROC_{1t} &= \alpha_{10} + \alpha_{11}ROC_{1,t-1} + \cdots + \alpha_{1k}ROC_{1,t-k} + \beta_{11}LR_{1,t-1} + \cdots + \beta_{1k}LR_{1,t-k} + \varepsilon_{1t} \\ LR_{2t} &= \alpha_{20} + \alpha_{21}LR_{2,t-1} + \cdots + \alpha_{2k}LR_{2,t-k} + \beta_{11}ROC_{2,t-1} + \cdots + \beta_{1k}ROC_{2,t-k} + \varepsilon_{2t} \end{aligned}$$

◆ 格兰杰因果检验

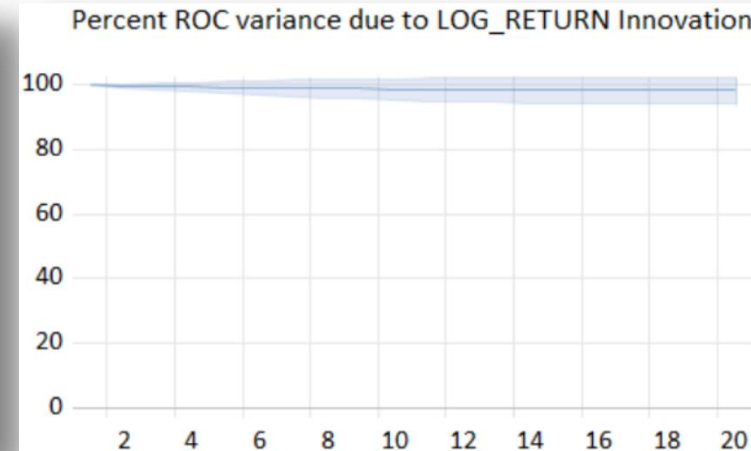
Pairwise Granger Causality Tests
Date: 12/21/25 Time: 11:54
Sample: 10/09/2020 9/30/2025
Lags: 10

Null Hypothesis:	Obs	F-Statistic	Prob.
ROC does not Granger Cause LOG_RETURN	1202	0.69741	0.7276
LOG_RETURN does not Granger Cause ROC		94.6026	3E-143

◆ 广义脉冲响应



◆ 方差分解





目录

研究背景

形态策略

技术指标

结论

3.3 实证结果——MFI

- 格兰杰因果检验：对数收益率是MFI的格兰杰原因，但MFI不是对数收益率的格兰杰原因
- 脉冲相应：MFI一个单位的正向冲击，对未来对数收益率无显著影响
- 方差分解：MFI的方差在很大程度上由股票收益率变化所解释

◆ VAR模型（根据信息准则确定滞后阶数为15）

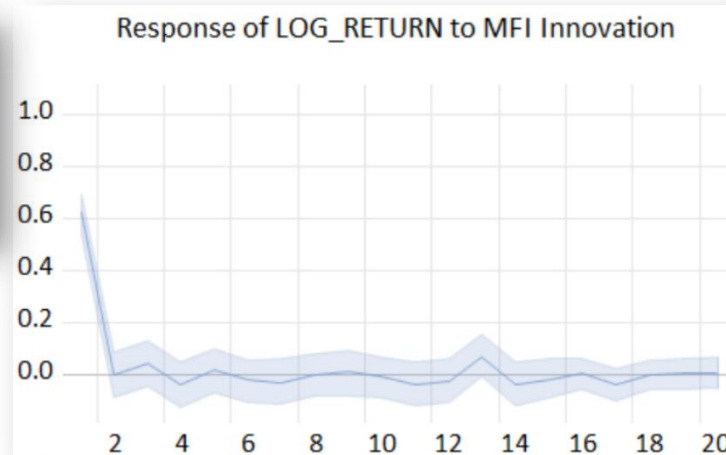
$$MFI_{1t} = \alpha_{10} + \alpha_{11}MFI_{1,t-1} + \cdots + \alpha_{1k}MFI_{1,t-k} + \beta_{11}LR_{1,t-1} + \cdots + \beta_{1k}LR_{1,t-k} + \varepsilon_{1t}$$
$$LR_{2t} = \alpha_{20} + \alpha_{21}LR_{2,t-1} + \cdots + \alpha_{2k}LR_{2,t-k} + \beta_{11}MFI_{2,t-1} + \cdots + \beta_{1k}MFI_{2,t-k} + \varepsilon_{2t}$$

◆ 格兰杰因果检验

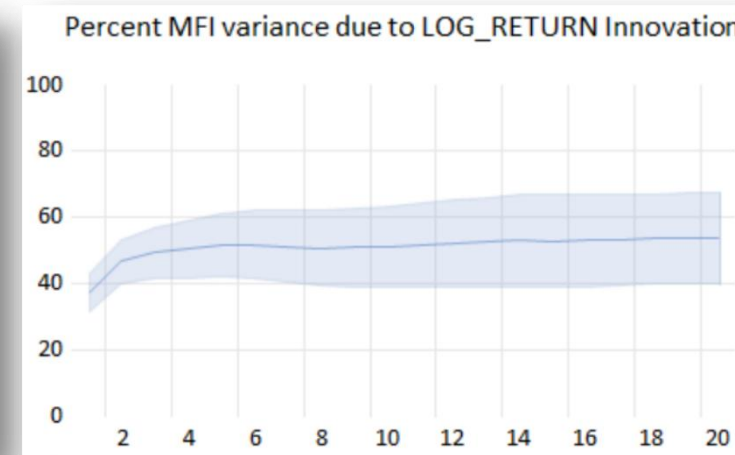
Pairwise Granger Causality Tests
Date: 12/21/25 Time: 11:55
Sample: 10/09/2020 9/30/2025
Lags: 15

Null Hypothesis:	Obs	F-Statistic	Prob.
MFI does not Granger Cause LOG_RETURN	1197	1.00252	0.4496
LOG_RETURN does not Granger Cause MFI		12.6089	1.E-29

◆ 广义脉冲响应



◆ 方差分解





目录

研究背景

形态策略

技术指标

结论

04 总结



总结与思考

目录

研究背景

形态策略

技术指标

结论

单纯基于K线图的经典形态策略无法持续战胜市场，严格遵循形态反而可能导致收益下滑。

缺乏成交量验证

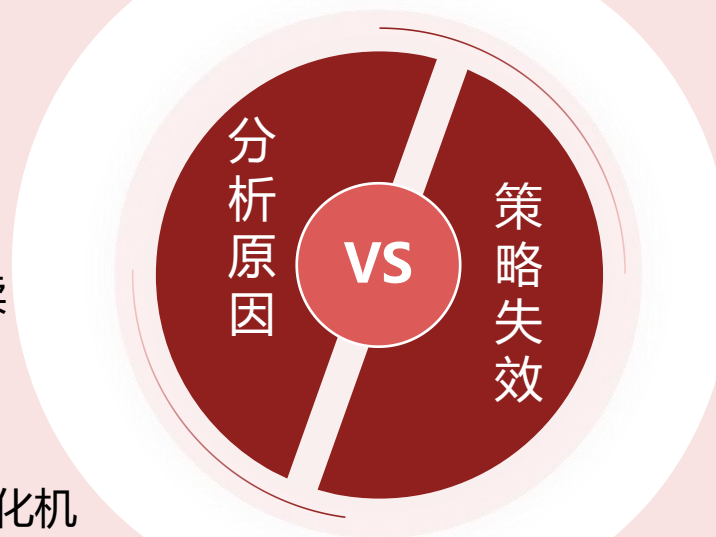
经典技术分析强调量价配合。我们的DTW模型仅基于价格计算形态相似度，没有考虑成交量的验证作用，导致模型可能被误导

未考虑技术指标

实战中，交易员会结合MACD、RSI等技术指标产生共振信号。仅靠图形的几何相似度作为唯一买卖依据，可能是其失效的原因

机构做局

使用的是最为经典的K线形态，量化机构和做市商非常清楚散户会在经典的形态突破点设置止损单或追涨单。机构可能故意制造出完美的突破形态诱导散户入场，随后迅速反向操作进行割韭菜



胜率低

越是严格符合教科书定义的形态，胜率反而越低，且未优于随机交易

跑输基准

策略收益显著低于基准收益(54.97%)，无论是在市场高波动期还是低波动期，该策略均未能产生稳定的超额收益



总结与思考

目录

研究背景

MACDI、RSI、ROC、MFI变化由历史收益率驱动，是价格的滞后反映，无法提供对未来收益率的预测信息

形态策略

格兰杰因果检验

MACDI、RSI、ROC、MFI 均不是股价对数收益率的格兰杰原因

脉冲响应分析

MFI与收益率之间主要是同期关联，而非领先滞后关系，未来收益率对各技术指标冲击的响应几乎为0

方差分解

技术指标波动的变化主要由收益率解释，表明技术指标很大程度上是股价信息的再综合，很少独立于价格之外的“新信息”

技术指标

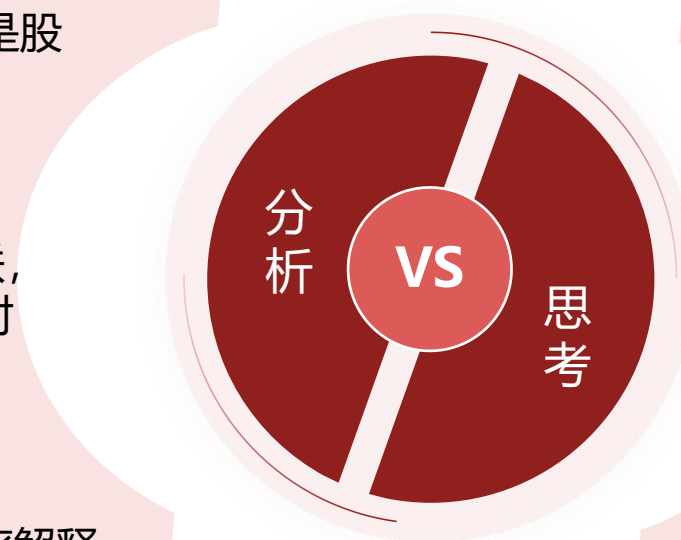
结论

交易策略

若基于技术指标构建交易策略，应注重**高频或日内**数据，避免基于日以上频次的滞后信号

市场有效性

市场在一定程度上具备信息效率，历史价格信息难以用于预测未来收益





南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

恳请老师同学批评指正!

 汇报小组：李根 鲜俊昊 甘政 王悯汐

 时间：2025.12.24



表 4: 高波动期 - 交易次数 (Transaction Frequency)

DTW阈值	Day 1	Day 3	Day 5	Day 7	Day 9	Day 11	Day 13	Day 15
0.0500	27.0	27.0	27.0	27.0	27.0	27.0	27.0	27.0
0.0600	63.0	63.0	63.0	63.0	63.0	63.0	63.0	63.0
0.0700	103.0	103.0	103.0	103.0	103.0	103.0	103.0	103.0
0.0800	156.0	156.0	156.0	156.0	156.0	156.0	156.0	156.0
0.0900	202.0	202.0	202.0	202.0	202.0	202.0	202.0	202.0
0.1000	252.0	252.0	252.0	252.0	252.0	252.0	252.0	252.0
0.1200	299.0	299.0	299.0	299.0	299.0	299.0	299.0	299.0
0.1400	328.0	328.0	328.0	328.0	328.0	328.0	328.0	328.0
0.1600	354.0	354.0	354.0	354.0	354.0	354.0	354.0	354.0
10.0	378.0	378.0	378.0	378.0	378.0	378.0	378.0	378.0

表 5: 高波动期 - 胜率 (Win Rate)

DTW阈值	Day 1	Day 3	Day 5	Day 7	Day 9	Day 11	Day 13	Day 15
0.0500	40.7	55.6	44.4	44.4	25.9	44.4	37.0	29.6
0.0600	49.2	47.6	52.4	54.0	47.6	50.8	47.6	54.0
0.0700	50.5	52.4	50.5	50.5	49.5	52.4	54.4	53.4
0.0800	57.7	53.9	49.4	51.3	46.8	52.6	55.1	53.2
0.0900	48.5	46.5	40.6	44.5	45.5	44.1	48.0	48.5
0.1000	51.2	49.6	44.8	45.6	48.0	43.6	47.6	47.6
0.1200	51.5	51.8	46.8	48.2	48.2	46.5	48.2	46.5
0.1400	51.2	52.4	49.1	47.9	48.5	45.7	47.0	45.4
0.1600	50.6	51.1	48.3	47.7	48.3	46.9	46.0	44.6
10.0	49.2	51.1	47.6	47.6	47.9	46.0	47.4	45.8

表 6: 高波动期 - 平均收益率 (Return Rate)

DTW阈值	Day 1	Day 3	Day 5	Day 7	Day 9	Day 11	Day 13	Day 15
0.0500	-1.0551	-0.9305	-1.4417	-2.1322	-3.3749	-3.4111	-4.2129	-5.5907
0.0600	0.3825	-0.2008	0.3046	0.0178	-0.2253	-0.8993	-1.4900	-1.6707
0.0700	0.4224	0.5407	0.8431	0.3759	0.4799	0.4542	-0.3057	-0.1341
0.0800	0.4497	0.1307	-0.0553	-0.2388	0.3807	0.5226	0.1126	0.4956
0.0900	-0.0841	-0.4546	-1.3133	-1.0598	-1.0985	-1.4104	-1.6154	-1.4115
0.1000	0.1162	-0.0986	-0.6146	-0.8115	-0.8393	-1.1107	-0.9812	-1.0830
0.1200	0.0970	-0.0255	-0.3838	-0.7308	-0.4578	-0.5666	-0.7059	-0.9863
0.1400	0.1060	0.1620	-0.1148	-0.6582	-0.5124	-0.7057	-0.8455	-0.8920
0.1600	0.1118	0.2058	-0.0305	-0.5830	-0.3344	-0.5092	-0.7149	-0.8257
10.0	0.0486	0.0660	-0.2338	-0.7343	-0.4982	-0.5262	-0.6400	-0.6708



表 7: 低波动期 - 交易次数 (Transaction Frequency)

DTW阈值	Day 1	Day 3	Day 5	Day 7	Day 9	Day 11	Day 13	Day 15
0.0500	3.0000	3.0000	3.0000	3.0000	3.0000	3.0000	3.0000	3.0000
0.0600	14.0	14.0	14.0	14.0	14.0	14.0	14.0	14.0
0.0700	40.0	40.0	40.0	40.0	40.0	40.0	40.0	40.0
0.0800	73.0	73.0	73.0	73.0	73.0	73.0	73.0	73.0
0.0900	118.0	118.0	118.0	118.0	118.0	118.0	118.0	118.0
0.1000	161.0	161.0	161.0	161.0	161.0	161.0	161.0	161.0
0.1200	193.0	193.0	193.0	193.0	193.0	193.0	193.0	193.0
0.1400	234.0	234.0	234.0	234.0	234.0	234.0	234.0	234.0
0.1600	255.0	255.0	255.0	255.0	255.0	255.0	255.0	255.0
10.0	267.0	267.0	267.0	267.0	267.0	267.0	267.0	267.0

表 8: 低波动期 - 胜率 (Win Rate)

DTW阈值	Day 1	Day 3	Day 5	Day 7	Day 9	Day 11	Day 13	Day 15
0.0500	0.0000	0.0000	33.3	66.7	66.7	33.3	33.3	33.3
0.0600	64.3	28.6	35.7	35.7	42.9	42.9	42.9	42.9
0.0700	45.0	50.0	57.5	52.5	52.5	52.5	47.5	52.5
0.0800	46.6	50.7	58.9	56.2	54.8	48.0	45.2	49.3
0.0900	49.1	50.0	52.5	51.7	51.7	46.6	45.8	47.5
0.1000	49.1	50.9	54.7	52.2	52.8	49.7	46.6	48.5
0.1200	50.8	52.3	53.9	51.3	52.9	48.7	46.1	48.7
0.1400	49.1	53.0	51.3	50.9	50.9	48.3	46.6	48.7
0.1600	51.8	53.7	52.5	52.2	50.6	48.2	46.3	48.6
10.0	49.1	53.2	50.9	50.2	48.7	46.8	47.2	49.4

表 9: 低波动期 - 平均收益率 (Return Rate)

DTW阈值	Day 1	Day 3	Day 5	Day 7	Day 9	Day 11	Day 13	Day 15
0.0500	-1.0439	-1.7393	-2.6941	-1.2140	-2.0537	-2.4867	-2.6070	-1.2921
0.0600	0.4535	-0.7079	-1.2188	-1.2483	-0.8832	-1.3460	-1.7900	-2.0010
0.0700	-0.0062	-0.0032	0.0368	-0.8089	-1.1366	-1.1579	-1.4678	-0.8166
0.0800	-0.1403	-0.0753	0.3567	-0.2252	-0.3614	-0.4380	-0.6970	-0.6360
0.0900	-0.1873	-0.0203	0.3999	-0.1714	-0.4109	-0.5893	-0.7746	-0.7463
0.1000	-0.0117	0.4107	0.8344	0.6089	0.5167	0.4506	0.0548	0.0055
0.1200	0.0521	0.4845	0.7377	0.5478	0.5757	0.4441	0.0021	-0.0723
0.1400	0.0202	0.4237	0.6092	0.2211	0.1454	0.0496	-0.1679	-0.1653
0.1600	0.0639	0.4193	0.6223	0.6380	0.6053	0.4482	0.1040	0.0899
10.0	-0.0156	0.3112	0.4273	0.2388	0.2807	0.3272	0.2516	0.1222

附录——平稳性检验



◆ 对数收益率

Null Hypothesis: LOG_RETURN has a unit root
Exogenous: None
Lag Length: 0 (Automatic - based on SIC, maxlag=22)

	t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic	-34.90569	0.0000
Test critical values: 1% level	-2.566873	
5% level	-1.941085	
10% level	-1.616523	

*MacKinnon (1996) one-sided p-values.

Null Hypothesis: RSI has a unit root
Exogenous: Constant
Lag Length: 14 (Automatic - based on SIC, maxlag=22)

	t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic	-6.106584	0.0000
Test critical values: 1% level	-3.435599	
5% level	-2.863746	
10% level	-2.567995	

*MacKinnon (1996) one-sided p-values.

Null Hypothesis: MFI has a unit root
Exogenous: Constant
Lag Length: 15 (Automatic - based on SIC, maxlag=22)

	t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic	-5.666170	0.0000
Test critical values: 1% level	-3.435604	
5% level	-2.863748	
10% level	-2.567996	

*MacKinnon (1996) one-sided p-values.

◆ RSI

◆ MFI

◆ MACDI

Null Hypothesis: MACDI has a unit root
Exogenous: None
Lag Length: 1 (Automatic - based on SIC, maxlag=22)

	t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic	-10.54886	0.0000
Test critical values: 1% level	-2.566875	
5% level	-1.941085	
10% level	-1.616523	

*MacKinnon (1996) one-sided p-values.

◆ ROC

Null Hypothesis: ROC has a unit root
Exogenous: None
Lag Length: 20 (Automatic - based on SIC, maxlag=22)

	t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic	-5.515384	0.0000
Test critical values: 1% level	-2.566905	
5% level	-1.941089	
10% level	-1.616520	

*MacKinnon (1996) one-sided p-values.

附录——LR&MACDI模型

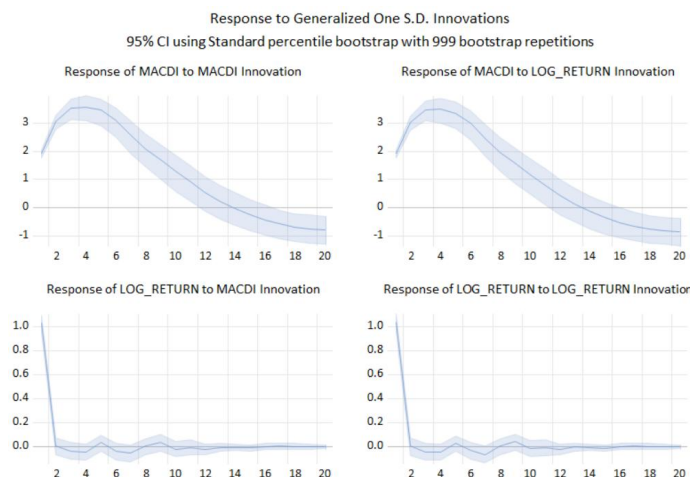


南方科技大学商学院
SUSTech Business School

◆ 信息准则确定滞后阶数

Lag	LogL	LR	FPE	AIC	SC	HQ
0	-3053.756	NA	86.43759	10.13518	10.14978	10.14086
1	-1874.036	2347.701	1.750417	6.235608	6.279408	6.252655
2	-1006.122	1721.434	0.099706	3.370222	3.443223	3.398634
3	-944.6623	121.4925	0.082405	3.179643	3.281843	3.219419
4	-908.4635	71.31711	0.074058	3.072847	3.204248	3.123988
5	-891.9215	32.48043	0.071041	3.031249	3.191849*	3.093753
6	-883.9787	15.54324	0.070118	3.018171	3.207972	3.092040*
7	-878.0733	11.51685	0.069677	3.011852	3.230853	3.097085
8	-873.8133	8.279815	0.069618	3.010990	3.259190	3.107588
9	-865.9794	15.17422	0.068739	2.998273	3.275674	3.106236
10	-856.0921	19.08598	0.067411*	2.978746*	3.285348	3.098074
11	-852.8393	6.257386	0.067579	2.981225	3.317026	3.111916
12	-850.2428	4.977712	0.067896	2.985880	3.350881	3.127936
13	-845.4667	9.124529	0.067723	2.983306	3.377507	3.136726
14	-842.4486	5.745825	0.067946	2.986563	3.409964	3.151348
15	-840.4115	3.864847	0.068392	2.993073	3.445675	3.169222
16	-839.5445	1.638957	0.069109	3.003464	3.485266	3.190978
17	-838.0188	2.874387	0.069681	3.011671	3.522673	3.210549
18	-835.7647	4.231643	0.070089	3.017462	3.557664	3.227704
19	-828.2526	14.05232*	0.069281	3.005813	3.575215	3.227420
20	-823.6815	8.520654	0.069154	3.003919	3.602521	3.236891

◆ 广义脉冲响应

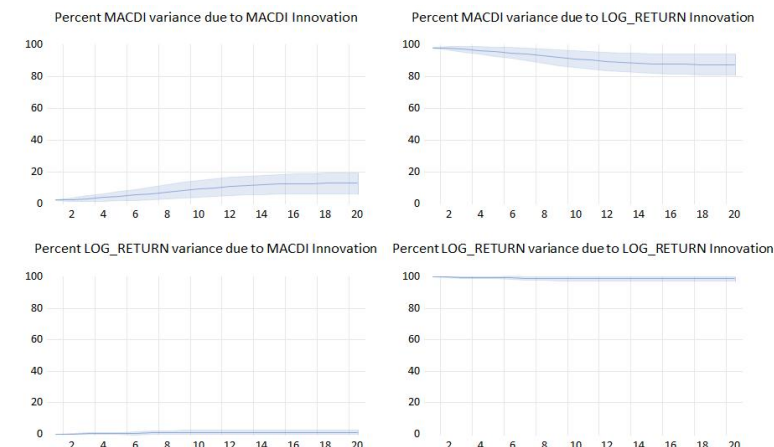


◆ VAR参数估计

Vector Autoregression Estimates		
Date: 12/21/25 Time: 11:50 Sample (adjusted): 10/23/2020 9/30/2025 Included observations: 1202 after adjustments Standard errors in () & t-statistics in []		
	LOG RETU	MACDI
LOG RETURN(-1)	0.137011 (0.19071) [0.71841]	0.050208 (0.35443) [0.14166]
LOG RETURN(-2)	-0.062540 (0.19076) [-0.32785]	-0.503448 (0.35451) [-1.42012]
LOG RETURN(-3)	0.140095 (0.19422) [0.72133]	0.143226 (0.36094) [0.39681]
LOG_RETURN(-4)	-0.171866 (0.19452) [-0.88354]	-0.686417 (0.36150) [-1.89879]
LOG_RETURN(-5)	0.439608 (0.19538) [2.25002]	0.851091 (0.36310) [2.34396]
LOG RETURN(-6)	-0.485970 (0.19428) [-2.50138]	-0.886592 (0.36106) [-2.45554]
LOG RETURN(-7)	-0.013039 (0.19151) [-0.06808]	-0.221105 (0.35591) [-0.62125]
LOG RETURN(-8)	0.075867 (0.19098) [0.39726]	-0.123981 (0.35492) [-0.34932]
LOG_RETURN(-9)	-0.007751 (0.05852) [-0.13246]	-0.048335 (0.10875) [-0.44446]
LOG_RETURN(-10)	-0.013015 (0.04344) [-0.29964]	-0.045605 (0.08072) [-0.56495]
MACDI(-1)	-0.071884 (0.10261) [-0.70059]	1.547424 (0.19069) [8.11506]
MACDI(-2)	0.155780 (0.20155) [0.77290]	-0.295838 (0.37457) [-0.78980]
MACDI(-3)	-0.201737 (0.21026) [-0.95945]	-0.582926 (0.39076) [-1.49177]
MACDI(-4)	0.286771 (0.21129) [1.35721]	0.770506 (0.39268) [1.96219]
MACDI(-5)	-0.491750 (0.21156) [-2.32435]	-1.222347 (0.39318) [-3.10889]
MACDI(-6)	0.692630	1.451287

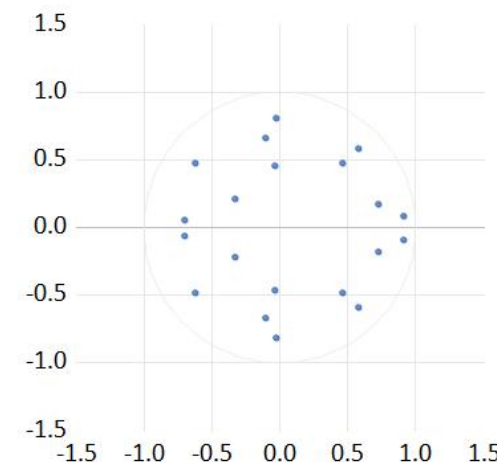
Vector Autoregression Estimates		
Date: 12/21/25 Time: 11:50 Sample (adjusted): 10/23/2020 9/30/2025 Included observations: 1202 after adjustments Standard errors in () & t-statistics in []		
MACDI(-7)	-0.514354 (0.21231) [-2.42261]	-0.857414 (0.39457) [-2.17302]
MACDI(-8)	0.105271 (0.21161) [0.49749]	0.161591 (0.39326) [0.41091]
MACDI(-9)	0.058444 (0.18511) [0.31573]	-0.024984 (0.34401) [-0.07263]
MACDI(-10)	-0.024409 (0.07224) [-0.33788]	0.023576 (0.13426) [0.17560]
C	-0.009363 (0.03208) [-0.29185]	-0.013893 (0.05962) [-0.23303]
R-squared	0.021873	0.961904
Adj. R-squared	0.005308	0.961259
Sum sq. resid	1452.126	5015.318
S.E. equation	1.108861	2.060745
F-statistic	1.320462	1491.001
Log likelihood	-1819.178	-2564.099
Akaike AIC	3.061861	4.301329
Schwarz SC	3.150818	4.390286
Mean dependent	-0.009971	0.004566
S.D. dependent	1.111816	10.46986
Determinant resid covariance (dof adj.)	0.119516	0.115377
Determinant resid covariance	0.115377	-2113.236
Log likelihood	-1819.178	-2564.099
Akaike information criterion	3.061861	4.301329
Schwarz criterion	3.150818	4.390286
Number of coefficients	42	

◆ 方差分解



◆ 模型平稳性检验

Inverse Roots of AR Characteristic Polynomial



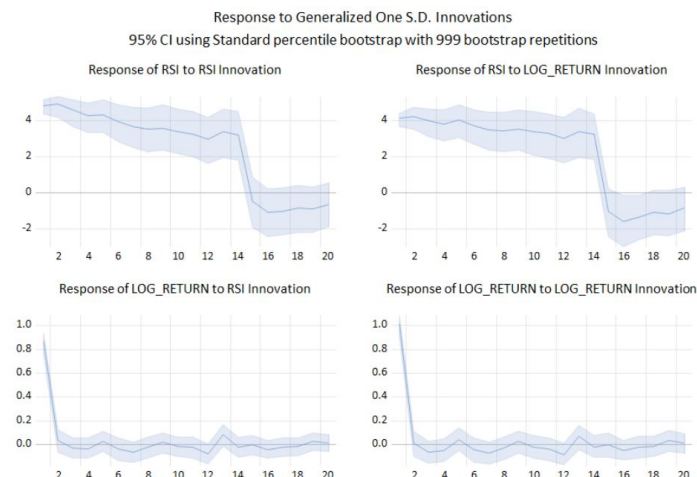
附录——LR&RSI模型



◆ 信息准则确定滞后阶数

Lag	LogL	LR	FPE	AIC	SC	HQ
0	-3341.299	NA	224.3327	11.08888	11.10348	11.09457
1	-2679.290	1317.430	25.29688	8.906435	8.950235	8.923482
2	-2677.394	3.760028	25.47406	8.913414	8.986415	8.941825
3	-2671.988	10.68764	25.35553	8.908749	9.010949	8.948525
4	-2665.798	12.19492	25.17210	8.901486	9.032886	8.952626
5	-2661.710	8.026111	25.16487	8.901195	9.061796	8.963700
6	-2657.669	7.907552	25.16159	8.901059	9.090860	8.974929
7	-2652.407	10.26334	25.05662	8.896872	9.115873	8.982105
8	-2647.376	9.778051	24.97133	8.893453	9.141654	8.990051
9	-2639.149	15.93568	24.62397	8.879433	9.156834	8.987395
10	-2632.269	13.27974	24.39027	8.869882	9.176483	8.989209
11	-2604.874	52.70095	22.56963	8.792285	9.128087	8.922977
12	-2566.037	74.45400	20.10725	8.676739	9.041741	8.818795
13	-2475.466	173.0314	15.08907	8.389605	8.783806	8.543025
14	-2226.029	474.8808	6.685544	7.575553	7.998955*	7.740338*
15	-2221.758	8.104227	6.679742	7.574652	8.027254	7.750801
16	-2221.007	1.419462	6.752370	7.585429	8.067231	7.772943
17	-2217.419	6.759268	6.761892	7.586796	8.097798	7.785674
18	-2206.333	20.81113*	6.605142*	7.563294*	8.103496	7.773537
19	-2203.987	4.388422	6.641829	7.568780	8.138182	7.790387
20	-2201.342	4.930399	6.672136	7.573274	8.171876	7.806246

◆ 广义脉冲响应



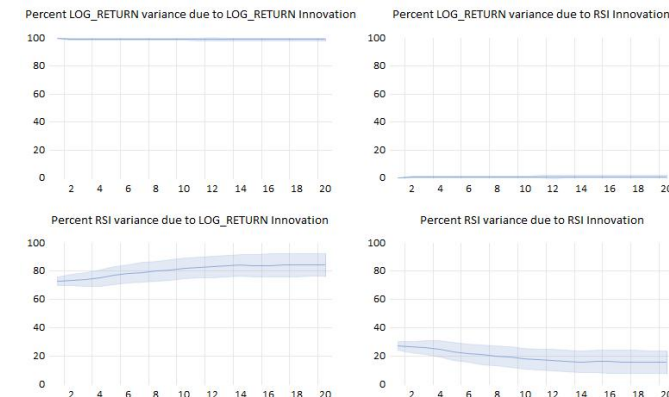
◆ VAR参数估计

Vector Autoregression Estimates		
Date: 12/21/25 Time: 17:16		
Sample (adjusted): 10/29/2020 9/30/2025		
Included observations: 1198 after adjustments		
Standard errors in () & t-statistics in []		
	LOG_RETU	RSI
LOG_RETURN(-1)	-0.054356 (0.03946) [-1.37761]	0.148152 (0.20294) [0.73002]
LOG_RETURN(-2)	-0.010346 (0.03944) [-0.26233]	0.349404 (0.20285) [1.72244]
LOG_RETURN(-3)	-0.039739 (0.03951) [-1.00590]	0.256675 (0.20319) [1.26320]
LOG_RETURN(-4)	0.034480 (0.03949) [0.87314]	0.395381 (0.20311) [1.94661]
LOG_RETURN(-5)	0.001001 (0.03938) [0.02542]	0.314911 (0.20253) [1.55491]
LOG_RETURN(-6)	-0.057728 (0.03928) [-1.46967]	0.039721 (0.20203) [0.19661]
LOG_RETURN(-7)	-0.066798 (0.03915) [-1.70631]	0.211516 (0.20135) [1.05048]
LOG_RETURN(-8)	0.010476 (0.03931) [0.26646]	0.445117 (0.20221) [2.20126]
LOG_RETURN(-9)	0.001340 (0.03929) [0.03410]	0.406753 (0.20208) [2.01284]
LOG_RETURN(-10)	-0.036027 (0.03933) [-0.91595]	0.276201 (0.20231) [1.36525]
LOG_RETURN(-11)	-0.046152 (0.03951) [-1.16816]	0.292924 (0.20321) [1.44149]
LOG_RETURN(-12)	0.008715 (0.03962) [0.21999]	0.776206 (0.20376) [3.80938]
LOG_RETURN(-13)	-0.035040 (0.04012) [-0.87340]	0.422365 (0.20635) [2.04684]
LOG_RETURN(-14)	-0.025009 (0.03466) [-0.72162]	-3.883371 (0.17826) [-21.7853]
RSI(-1)	0.012467 (0.00647) [1.92669]	0.964756 (0.03328) [28.9883]
RSI(-2)	-0.003886	-0.073384

Vector Autoregression Estimates

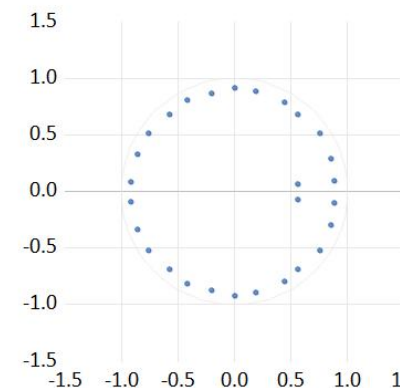
	(0.00767) [-0.50695]	(0.03943) [-1.86114]
RSI(-3)	-0.005312 (0.00767) [-0.69246]	0.000318 (0.03946) [0.00807]
RSI(-4)	0.001993 (0.00767) [0.25989]	0.037272 (0.03945) [0.94477]
RSI(-5)	-0.010905 (0.00769) [-1.41746]	-0.079988 (0.03957) [-2.02135]
RSI(-6)	0.005503 (0.00771) [0.71395]	0.067802 (0.03964) [1.71027]
RSI(-7)	0.013156 (0.00771) [1.70740]	0.023289 (0.03963) [0.58761]
RSI(-8)	-0.013385 (0.00771) [-1.73493]	-0.056636 (0.03968) [-1.42729]
RSI(-9)	-0.001868 (0.00771) [-0.24213]	0.002950 (0.03967) [0.07436]
RSI(-10)	0.008710 (0.00769) [1.13321]	0.018751 (0.03953) [0.47430]
RSI(-11)	-0.005638 (0.00769) [-0.73347]	-0.018784 (0.03954) [-0.47511]
RSI(-12)	0.006885 (0.00769) [0.89560]	0.017286 (0.03954) [0.43721]
RSI(-13)	-0.005907 (0.00768) [-0.76894]	-0.024067 (0.03951) [-0.60912]
RSI(-14)	0.000845 (0.00546) [0.15459]	0.013978 (0.02811) [0.49733]
C	-0.142022 (0.23374) [-0.60761]	5.230135 (1.20222) [4.35041]
R-squared	0.025173	0.894233
Adj. R-squared	0.001824	0.891700
Sum sq. resids	1444.826	38222.69
S.E. equation	1.111733	5.718121
F-statistic	1.078133	352.9860
Log likelihood	-1812.102	-3774.091
Akaike AIC	3.073626	6.349067
Schwarz SC	3.196802	6.472242
Mean dependent	-0.008598	48.98347
S.D. dependent	1.112748	17.37555
Determinant resid covariance		15.64644
Determinant resid covariance (dof adj.)		14.89810
Log likelihood		-5017.816

◆ 方差分解



◆ 模型平稳性检验

Inverse Roots of AR Characteristic Polynomial



Vector Autoregression Estimates

Akaike information criterion	8.473816
Schwarz criterion	8.720166
Number of coefficients	58

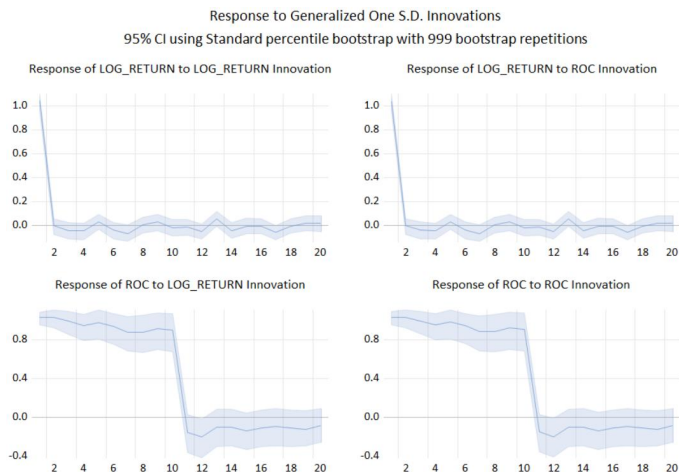
附录——LR&ROC模型



◆ 信息准则确定滞后阶数

Lag	LogL	LR	FPE	AIC	SC	HQ
0	-2349.388	NA	8.357929	7.798965	7.813565	7.804647
1	-1742.343	1208.050	1.130953	5.798815	5.842615	5.815861
2	-1736.773	11.04724	1.125080	5.793608	5.866608	5.822019
3	-1726.920	19.47695	1.103452	5.774195	5.876396	5.813971
4	-1711.589	30.20471	1.062753	5.736613	5.868013	5.787753
5	-1696.383	29.85661	1.023983	5.699446	5.860047	5.761951
6	-1687.288	17.79816	1.006829	5.682547	5.872348	5.756416
7	-1651.980	68.85837	0.907533	5.578708	5.797709	5.663941
8	-1557.819	183.0135	0.672967	5.279665	5.527866	5.376263
9	-115.5677	2793.615	0.005705	0.509346	0.786747	0.617308
10	78.33458	374.2989	0.003039	-0.120513	0.186088*	-0.001186*
11	82.55592	8.120653	0.003037*	-0.121247*	0.214554	0.009445
12	83.09342	1.030433	0.003072	-0.109763	0.255239	0.032293
13	86.34480	6.211587	0.003080	-0.107280	0.286922	0.046141
14	89.44967	5.911100	0.003089	-0.104311	0.319091	0.060474
15	91.61673	4.111299	0.003108	-0.098231	0.354370	0.077918
16	93.37926	3.332150	0.003131	-0.090810	0.390992	0.096704
17	101.0068	14.36969	0.003094	-0.102842	0.408160	0.096036
18	105.8184	9.032589	0.003086	-0.105534	0.434689	0.104709
19	112.4175	12.34473*	0.003059	-0.114154	0.455248	0.107453
20	113.4426	1.910699	0.003090	-0.104287	0.494315	0.128685

◆ 广义脉冲响应



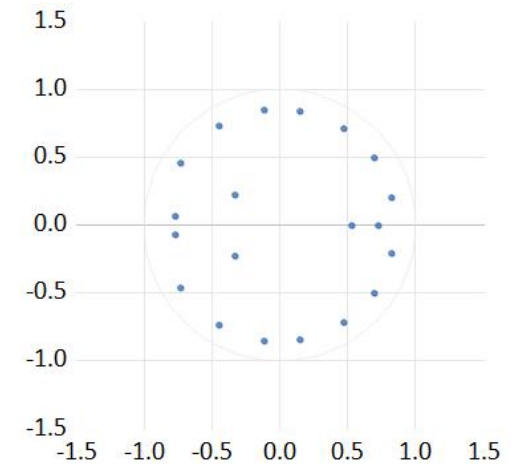
◆ VAR参数估计

Vector Autoregression Estimates		
Date: 12/21/25 Time: 11:54 Sample (adjusted): 10/23/2020 9/30/2025 Included observations: 1202 after adjustments Standard errors in () & t-statistics in []		
	LOG RETU	ROC
LOG RETURN(-1)	-0.061364 (0.19270) [-0.31845]	0.102715 (0.19616) [0.52362]
LOG RETURN(-2)	0.041239 (0.19257) [0.21415]	0.215887 (0.19604) [1.10126]
LOG RETURN(-3)	-0.069104 (0.19412) [-0.35599]	0.096876 (0.19761) [0.49024]
LOG_RETURN(-4)	0.023753 (0.19416) [0.12234]	0.193594 (0.19765) [0.97946]
LOG_RETURN(-5)	-0.054312 (0.19465) [-0.27902]	0.112336 (0.19815) [0.56691]
LOG RETURN(-6)	-0.120230 (0.19457) [-0.61793]	0.038300 (0.19807) [0.19337]
LOG RETURN(-7)	-0.025178 (0.19333) [-0.13024]	0.136220 (0.19681) [0.69215]
LOG RETURN(-8)	-0.001460 (0.19245) [-0.00758]	0.161769 (0.19591) [0.82573]
LOG_RETURN(-9)	-0.021947 (0.19198) [-0.11431]	0.139770 (0.19544) [0.71516]
LOG_RETURN(-10)	-0.027162 (0.18984) [-0.14308]	-0.869885 (0.19325) [-4.50132]
ROC(-1)	0.056995 (0.18769) [0.30367]	0.897609 (0.19106) [4.69799]
ROC(-2)	-0.076630 (0.04079) [-1.87852]	-0.079965 (0.04153) [-1.92562]
ROC(-3)	0.058481 (0.04084) [1.43201]	0.058989 (0.04157) [1.41893]
ROC(-4)	-0.009857 (0.04086) [-0.24122]	-0.008069 (0.04160) [-0.19397]
ROC(-5)	0.001372 (0.04082) [0.03360]	-0.000844 (0.04155) [-0.02031]
ROC(-6)	0.028516	0.028130

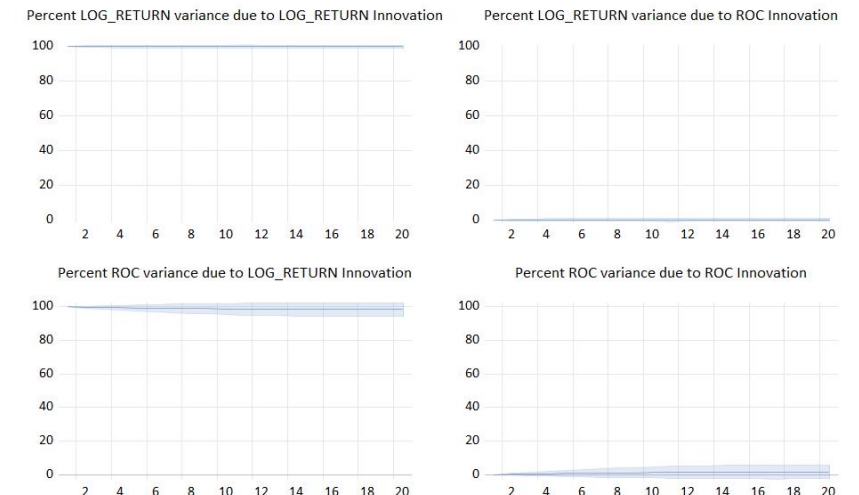
Vector Autoregression Estimates		
Date: 12/21/25 Time: 11:54 Sample (adjusted): 10/23/2020 9/30/2025 Included observations: 1202 after adjustments Standard errors in () & t-statistics in []		
ROC(-7)	(0.04072) [0.70022]	(0.04146) [0.67854]
ROC(-8)	-0.050851 (0.04063) [-1.25143]	-0.048267 (0.04137) [-1.16684]
ROC(-9)	0.003575 (0.04065) [0.08794]	-0.000448 (0.04138) [-0.01083]
ROC(-10)	-0.001667 (0.04059) [-0.04106]	0.000276 (0.04132) [0.00669]
C	0.005617 (0.02862) [0.19624]	0.005061 (0.02914) [0.17369]
R-squared	0.014940	0.899531
Adj. R-squared	-0.001742	0.897830
Sum sq. resid	1462.419	1515.516
S.E. equation	1.112784	1.132806
F-statistic	0.895559	528.6946
Log likelihood	-1823.423	-1844.858
Akaike AIC	3.068924	3.104588
Schwarz SC	3.157881	3.193545
Mean dependent	-0.009971	-0.051066
S.D. dependent	1.111816	3.543996
Determinant resid covariance (dof adj.)		0.011285
Determinant resid covariance		0.010894
Log likelihood		-694.8707
Akaike information criterion		1.226074
Schwarz criterion		1.403989
Number of coefficients		42

◆ 模型平稳性检验

Inverse Roots of AR Characteristic Polynomial



◆ 方差分解



附录——LR&MFI模型

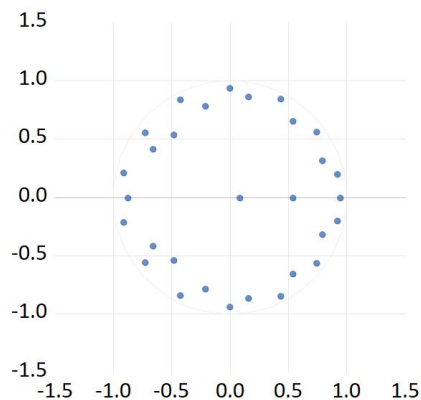


◆ 信息准则确定滞后阶数

Lag	LogL	LR	FPE	AIC	SC	HQ
0	-3332.403	NA	217.8107	11.05938	11.07398	11.06506
1	-2613.752	1430.152	20.35461	8.689061	8.732861	8.706108
2	-2608.414	10.58764	20.26448	8.684623	8.757623	8.713034
3	-2605.775	5.217447	20.35617	8.689136	8.791336	8.728912
4	-2599.538	12.28607	20.20580	8.681719	8.813120	8.732860
5	-2597.862	3.292370	20.36217	8.689425	8.850026	8.751930
6	-2595.856	3.925282	20.49740	8.696039	8.885840	8.769908
7	-2593.445	4.702198	20.60586	8.701309	8.920310	8.786543
8	-2591.981	2.845238	20.78011	8.709721	8.957922	8.806319
9	-2589.641	4.531805	20.89511	8.715228	8.992629	8.823190
10	-2586.188	6.666544	20.93333	8.717040	9.023641	8.836367
11	-2570.111	30.92708	20.11176	8.676985	9.012786	8.807676
12	-2562.532	14.52918	19.87486	8.665115	9.030116	8.807171
13	-2545.246	33.02336	19.01856	8.621049	9.015251	8.774470
14	-2459.113	163.9815	14.48377	8.348634	8.772036	8.513419
15	-2404.760	103.1174*	12.25647*	8.181626*	8.634228*	8.357776*
16	-2403.378	2.613502	12.36380	8.190308	8.672110	8.377822
17	-2400.902	4.663959	12.42699	8.195364	8.706366	8.394242
18	-2398.881	3.794383	12.50942	8.201927	8.742129	8.412170
19	-2398.527	0.662609	12.66227	8.214019	8.783421	8.435627
20	-2394.590	7.339093	12.66565	8.214227	8.812830	8.447199

◆ 模型平稳性检验

Inverse Roots of AR Characteristic Polynomial



◆ VAR参数估计

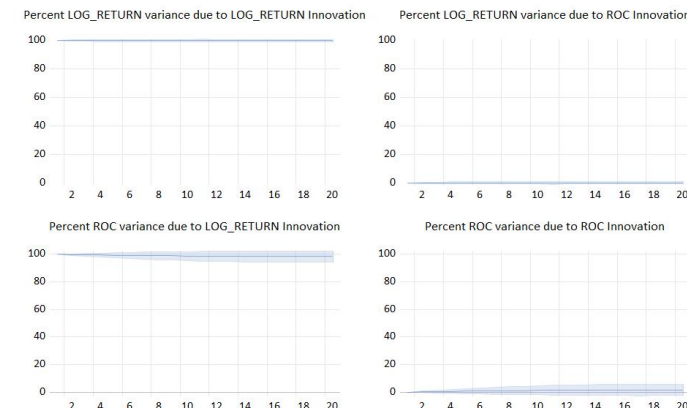
Vector Autoregression Estimates

Vector Autoregression Estimates Date: 12/21/25 Time: 17:16 Sample (adjusted): 10/30/2020 9/30/2025 Included observations: 1197 after adjustments Standard errors in () & t-statistics in []		
	LOG_RETU	MFI
LOG_RETURN(-1)	-0.001771 (0.03403) [-0.05206]	1.350061 (0.18909) [7.13994]
LOG_RETURN(-2)	0.001083 (0.03394) [0.03191]	0.576715 (0.18860) [3.05794]
LOG_RETURN(-3)	-0.037074 (0.03411) [-1.08704]	0.246424 (0.18953) [1.30016]
LOG_RETURN(-4)	0.052023 (0.03417) [1.52227]	0.310160 (0.18992) [1.63313]
LOG_RETURN(-5)	-0.010905 (0.03408) [-0.32001]	0.098860 (0.18937) [0.52204]
LOG_RETURN(-6)	-0.050086 (0.03403) [-1.47175]	-0.114795 (0.18912) [-0.60699]
LOG_RETURN(-7)	-0.030847 (0.03399) [-0.90744]	0.162091 (0.18891) [0.85803]
LOG_RETURN(-8)	0.003167 (0.03399) [0.09319]	0.277680 (0.18887) [1.47025]
LOG_RETURN(-9)	-0.003644 (0.03397) [-0.10725]	0.348870 (0.18880) [1.84780]
LOG_RETURN(-10)	0.016285 (0.03399) [0.47906]	0.506127 (0.18891) [2.67917]
LOG_RETURN(-11)	-0.051888 (0.03410) [-1.52162]	0.369217 (0.18951) [1.94832]
LOG_RETURN(-12)	0.031296 (0.03425) [0.91383]	0.310713 (0.19032) [1.63258]
LOG_RETURN(-13)	-0.012350 (0.03420) [-0.36112]	0.509312 (0.19005) [2.67995]
LOG_RETURN(-14)	0.011131 (0.03427) [0.32484]	-2.097493 (0.19043) [-11.0148]
LOG_RETURN(-15)	-0.013511 (0.03505) [-0.38547]	-0.746633 (0.19478) [-3.83318]
MFI(-1)	-0.001147	0.903192

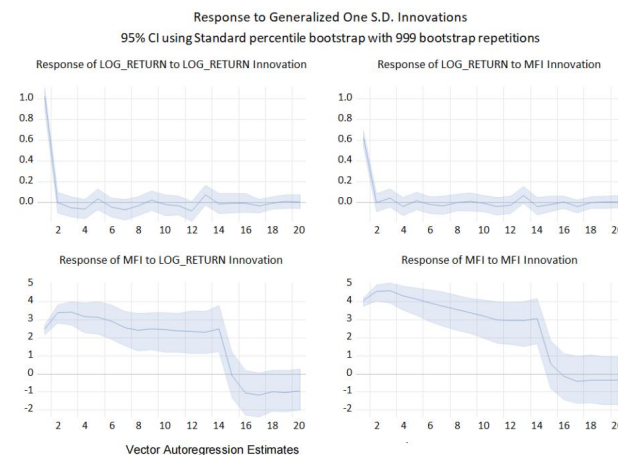
Vector Autoregression Estimates

	(0.00593) [-0.19342]	(0.03296) [27.3999]
MFI(-2)	0.010065 (0.00783) [1.28576]	0.023992 (0.04350) [0.55149]
MFI(-3)	-0.007948 (0.00711) [-1.11726]	-0.046109 (0.03953) [-1.16629]
MFI(-4)	-0.003261 (0.00709) [-0.45986]	0.043188 (0.03940) [1.09608]
MFI(-5)	-0.001378 (0.00708) [-0.19466]	-0.005451 (0.03935) [-0.13851]
MFI(-6)	0.002038 (0.00708) [0.28791]	0.036263 (0.03933) [0.92204]
MFI(-7)	0.008199 (0.00707) [1.15922]	0.010498 (0.03931) [0.26708]
MFI(-8)	-0.006297 (0.00707) [-0.89014]	-0.084367 (0.03931) [-2.14615]
MFI(-9)	-0.004229 (0.00708) [-0.59741]	0.014218 (0.03934) [0.36138]
MFI(-10)	-0.002726 (0.00707) [-0.38571]	-0.025492 (0.03928) [-0.64903]
MFI(-11)	0.013440 (0.00706) [1.90231]	0.060969 (0.03926) [1.55292]
MFI(-12)	-0.007470 (0.00706) [-1.05738]	-0.013074 (0.03926) [-0.33301]
MFI(-13)	-0.004380 (0.00706) [-0.62011]	0.006382 (0.03925) [0.16257]
MFI(-14)	-0.001225 (0.00706) [-0.17355]	-0.217888 (0.03924) [-5.55300]
MFI(-15)	0.005136 (0.00499) [1.02919]	0.201590 (0.02773) [7.26923]
C	0.052437 (0.15938) [0.32901]	4.853917 (0.88569) [5.48040]
R-squared	0.024751	0.903710
Adj. R-squared	-0.000341	0.901233
Sum sq. resids	1445.335	44635.99
S.E. equation	1.113358	6.187188
F-statistic	0.986396	364.7749
Log likelihood	-1811.300	-3864.275

◆ 方差分解



◆ 广义脉冲响应



Vector Autoregression Estimates		
Akaike AIC	3.078196	6.508395
Schwarz SC	3.209954	6.640154
Mean dependent	-0.008887	52.47537
S.D. dependent	1.113168	19.68730
Determinant resid covariance (dof adj.)		34.73303
Determinant resid covariance		32.95729
Log likelihood		-5488.823
Akaike information criterion		9.274559
Schwarz criterion		9.538076
Number of coefficients		62